

Killara HIGH SCHOOL

MATHEMATICS DEPARTMENT

2020

Year 12

HSC Mathematics Extension 1

TASK 2

Date issued: 26/02/20

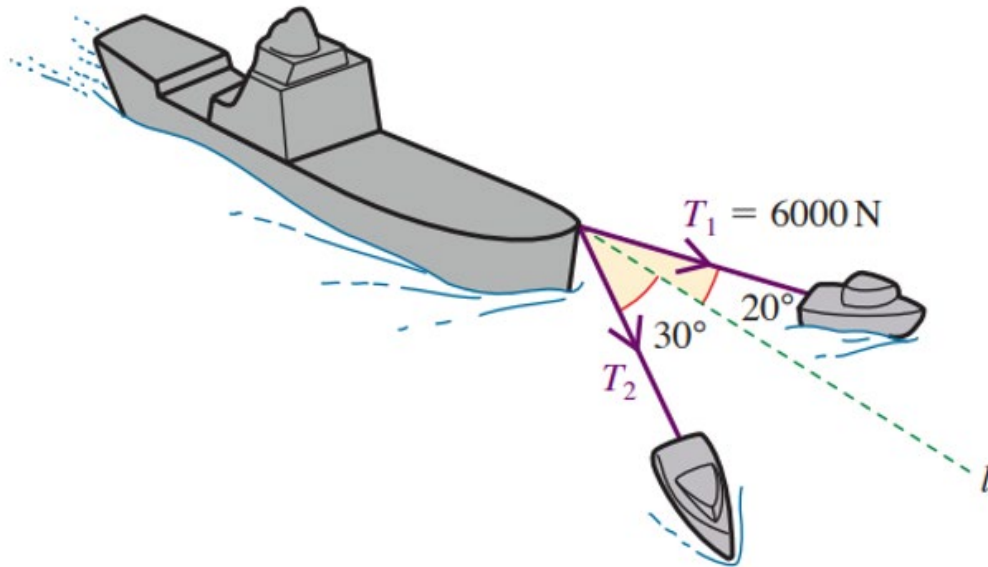
Date due: 16/03/2020

LEARNING OUTCOMES:

- Applies concepts and techniques involving vectors and projectiles to solve problems (ME12-2)
- Chooses and uses appropriate technology to solve problems in a range of contexts (ME12-6)
- Evaluates and justifies conclusions, communicating in a range of contexts (ME12-7)

Section I 40%	Part A – Modelling	/42
	Part B – Geometry	/23
	Part C – Modelling Orbits	/10
Section II 60%	35-minute examination	

- 1 A ship is being towed by two tugs. Each tug exerts forces on the ship as indicated. There is also a drag force on the ship.



- (a) Write down the components of the tensions in the towing cables along and perpendicular to the line of motion, l , of the ship. 2

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- (b) There is no resultant force perpendicular to the line l . Find T_2 . 2

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- (c) The ship is traveling at a constant speed. Find the magnitude of the drag force. 2

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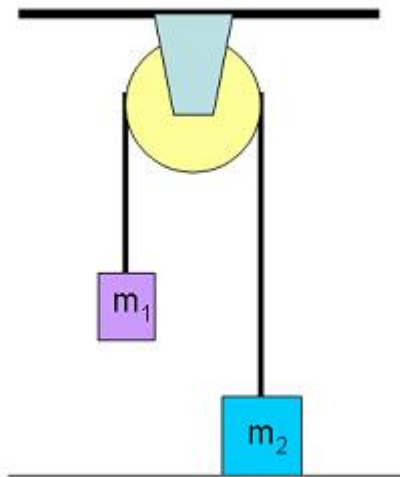
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- 2 Two blocks with masses m_1 and m_2 are connected by a cord that passes over a massless and frictionless pulley.



Find the mass of m_2 if $m_1=2\text{kg}$ and is accelerating upwards at 1.3ms^{-2}

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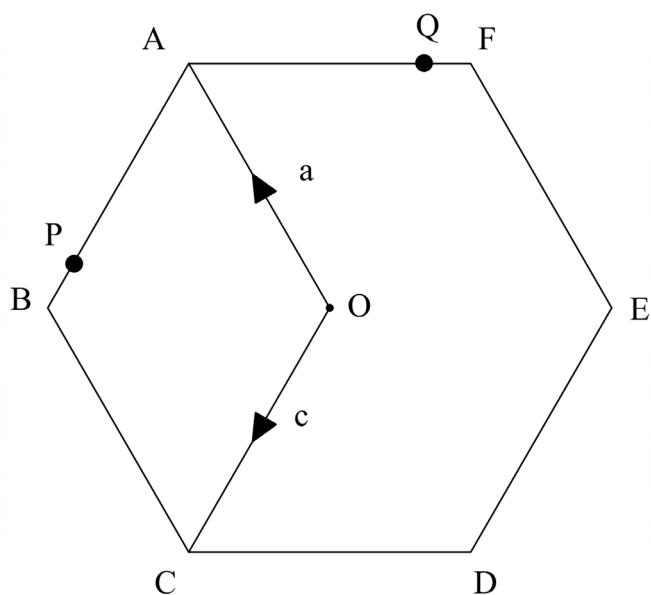
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- 3 $ABCDEF$ is a regular hexagon, centre O . $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OC} = \mathbf{c}$. P is a point on BA such that $BP:BA = 1:3$ and Q is a point on AF such that $AQ:QF = 2:1$.



- (a) Express, as simply as possible, in terms of $\mathbf{a}$ and/or $\mathbf{c}$.

(i) $\overrightarrow{AF}$ 1

(ii) $\overrightarrow{OQ}$ 2

(iii) $\overrightarrow{PA}$ 1

(iv) $\overrightarrow{BF}$ 1

(v) $\overrightarrow{PQ}$ 1

(b) Show that $\triangle APQ$ is similar to $\triangle APQ$.

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(c) PQ intersects OA at the point R . Find $\overrightarrow{PR}$.

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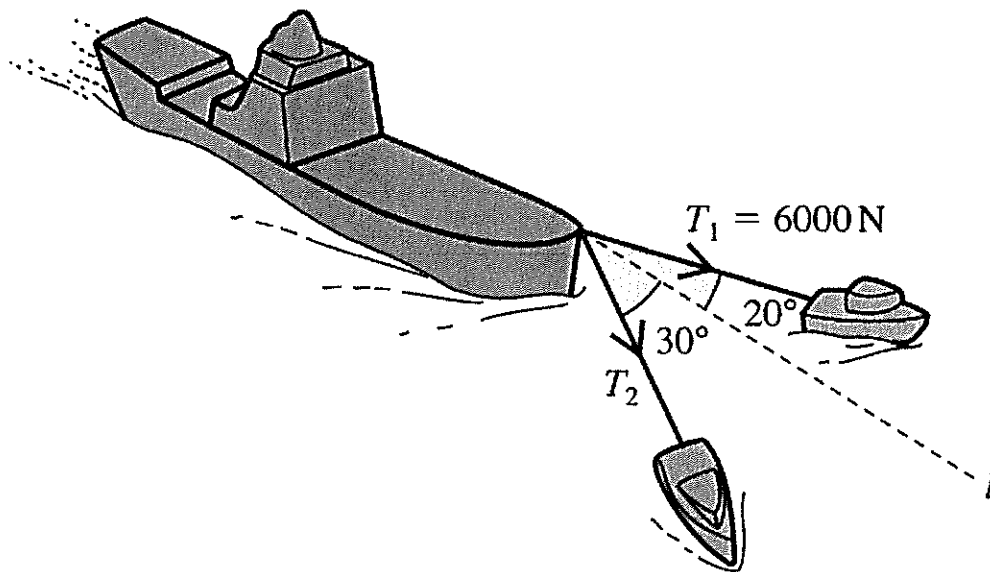
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A ship is being towed by two tugs. Each tug exerts forces on the ship as indicated. There is also a drag force on the ship.



- (i) Write down the components of the tensions in the towing cables along and perpendicular to the line of motion, l , of the ship.

① Missing a component or small error

$$T_1 = \begin{pmatrix} 6000 \cos 20^\circ \\ 6000 \sin 20^\circ \end{pmatrix} \quad T_2 = \begin{pmatrix} T_2 \cos 30^\circ \\ -T_2 \sin 30^\circ \end{pmatrix}$$

② All components in same form (does not have to be column vectors) (ii)

(Defining the vector as $\begin{pmatrix} \text{Along} \\ \text{perpendicular} \end{pmatrix}$)

There is no resultant force perpendicular to the line l . Find T_2 .

$$F_{\text{perpendicular}} = 6000 \sin 20^\circ - T_2 \sin 30^\circ = 0$$

— 0 mark

$$6000 \sin 20^\circ = T_2 \sin 30^\circ$$

$$T_2 = \frac{6000 \sin 20^\circ}{\sin 30^\circ}$$

$$= 4104 \text{ N}$$

← 0 mark

- (ii) The ship is traveling at a constant speed. Find the magnitude of the drag force.

$$F_{\text{Along}} = 6000 \cos 20^\circ + 4104 \cos 30^\circ - D = 0$$

1 mark ↗

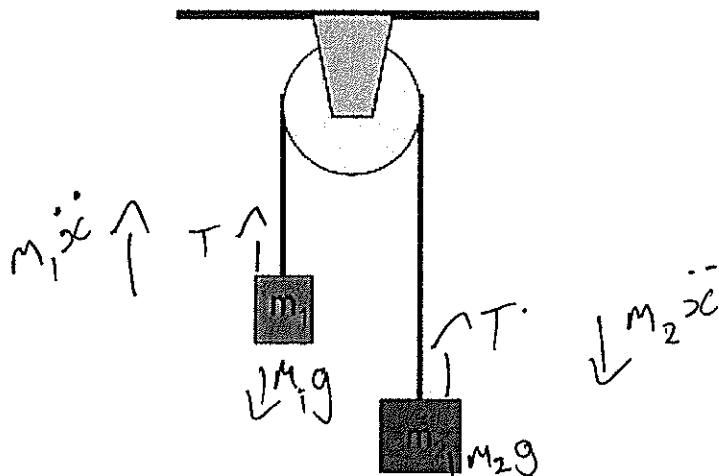
$$D = 6000 \cos 20^\circ + 4104 \cos 30^\circ$$

$$= 9193 \text{ N (9192 N)}$$

1 mark ↗

↑
If they use
4104

Two blocks with masses m_1 and m_2 are connected by a cord that passes over a massless and frictionless pulley.



① mark setting up vectors / vector diagrams

Find the mass of m_2 if $m_1 = 2\text{kg}$ and is accelerating upwards at 1.3ms^{-2}

$$2 \times 1.3 = T - m_1 g$$

$$= T - 2 \times 9.8 \quad \text{--- (1)}$$

$$m_2 \times 1.3 = m_2 \times 9.8 - T \quad \text{--- (2)}$$

① + ②

$$2 \times 1.3 + m_2 \times 1.3 = m_2 \times 9.8 - 2 \times 9.8$$

← 1 mark sum of forces in some some

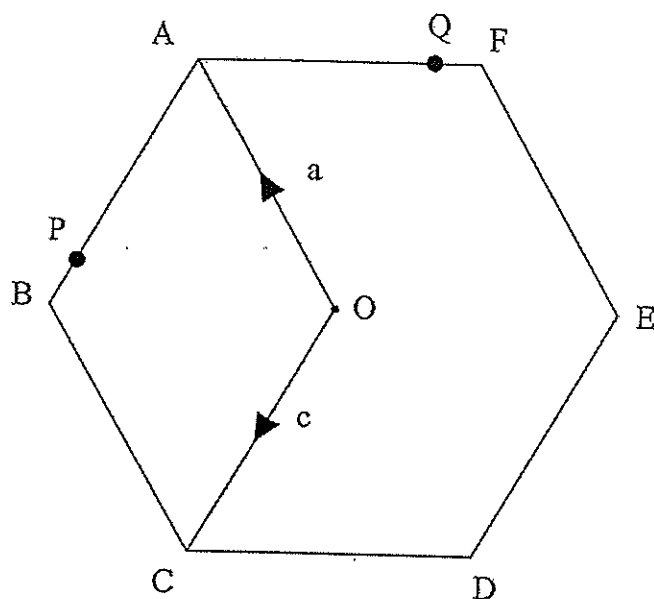
$$2 \times 11.1 = 8.5 m_2$$

$$m_2 = \frac{22.2}{8.5}$$

$$\approx 2.6 \text{ (I.D.P.)}$$

1 mark answer.

- 3 $ABCDEF$ is a regular hexagon, centre O . $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OC} = \mathbf{c}$. P is a point on BA such that $BP:BA = 1:3$ and Q is a point on AF such that $AQ:QF = 2:1$.



- (a) Express, as simply as possible, in terms of $\mathbf{a}$ and/or $\mathbf{c}$.

(i) $\overrightarrow{AF} = \overrightarrow{AO} + \overrightarrow{OF} = -\mathbf{a} - \mathbf{c}$

(ii) $\overrightarrow{OQ} = \overrightarrow{OA} + \overrightarrow{AQ} = \mathbf{a} - \frac{2}{3}\mathbf{a} - \frac{2}{3}\mathbf{c}$
 $= \frac{1}{3}\mathbf{a} - \frac{2}{3}\mathbf{c}$

(iii) $\overrightarrow{PA} = \frac{2}{3}\overrightarrow{BA} = -\frac{2}{3}\mathbf{c}$

(iv) $\overrightarrow{BF} = \overrightarrow{BA} + \overrightarrow{AF}$
 $= -\mathbf{c} - \mathbf{a} - \mathbf{c}$
 $= -2\mathbf{c} - \mathbf{a}$

(v) $\overrightarrow{PQ} = \overrightarrow{PA} + \overrightarrow{AQ} = -\frac{2}{3}\mathbf{c} - \frac{2}{3}\mathbf{a} - \frac{1}{3}\mathbf{c}$
 $= -\frac{2}{3}\mathbf{a} - \frac{4}{3}\mathbf{c}$

- (b) Show that $\triangle APQ$ is similar to $\triangle ABF$.

$$\vec{PQ} = \frac{2}{3}(-2\hat{c} - \hat{a}) = \frac{2}{3}\vec{BF}$$

$$\therefore PQ \parallel BF$$

$$\angle PAQ = \angle BAF \text{ (common } \angle)$$

$$\angle APQ = \angle ABF \text{ (corresponding angles in parallel lines)}$$

$$\therefore \triangle APQ \parallel \triangle ABF \text{ (equiangular)}$$

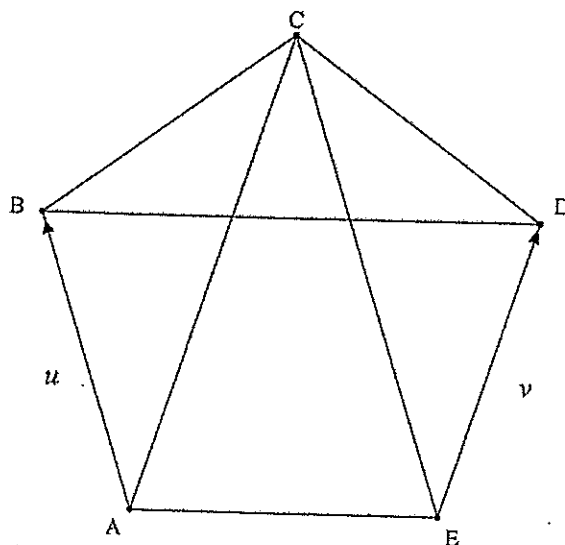
- (c) PQ intersects OA at the point R . Find $\vec{PR}$.

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$$\vec{PR} = \vec{PA} + \vec{AR} = -\frac{2}{3}\hat{a} + \frac{2}{3}\left(-\frac{1}{2}\hat{a}\right)$$

$$= -\frac{1}{3}\hat{a} - \frac{2}{3}\hat{a}$$

- 4 Consider the following pentagon $ABCDE$ with 5 equal lengths.



Let $u = \overrightarrow{AB}$ and $v = \overrightarrow{ED}$.

Observe that $\overrightarrow{BD}$ is parallel to $\overrightarrow{AE}$ and hence $\overrightarrow{BD} = \lambda(\overrightarrow{AE})$ for some $\lambda > 1$.

Similarly $\overrightarrow{AC} = \lambda(\overrightarrow{ED}) = \lambda v$ and $\overrightarrow{EC} = \lambda(\overrightarrow{AB}) = \lambda u$.

- (a) By considering the triangle ACE prove that $\overrightarrow{AE} = \lambda(v - u)$.

$$\overrightarrow{AE} = \lambda v - \lambda u = \lambda(v - u) \quad (1)$$

- (b) Find a different expression for $\overrightarrow{AE}$ in terms of λ , v and u and hence find the exact value of λ , expressing your answer in surd form.

$$\overrightarrow{AE} = u + \lambda \overrightarrow{AE} - v$$

$$\overrightarrow{AE} (1 - \lambda) = u - v$$

$$\overrightarrow{AE} = \frac{1}{1 - \lambda} (u - v)$$

$$\therefore \lambda = \frac{1}{\lambda - 1}$$

— (1) for λ quadratic

$$\lambda^2 - \lambda - 1 = 0$$

$$(\lambda > 0)$$

$$\lambda = \frac{\sqrt{5} + 1}{2}$$

(Golden Ratio)

— (1) for positive solution